Exploitation Lab Report

The objective of this phase is to simulate exploitation techniques on a vulnerable system to understand how attackers gain unauthorized access. The activity focuses on identifying exploitable services and using exploitation frameworks to gain system access. Tools such as **Metasploit Framework** are used to simulate real-world attack scenarios in a controlled laboratory environment.

Environment Setup

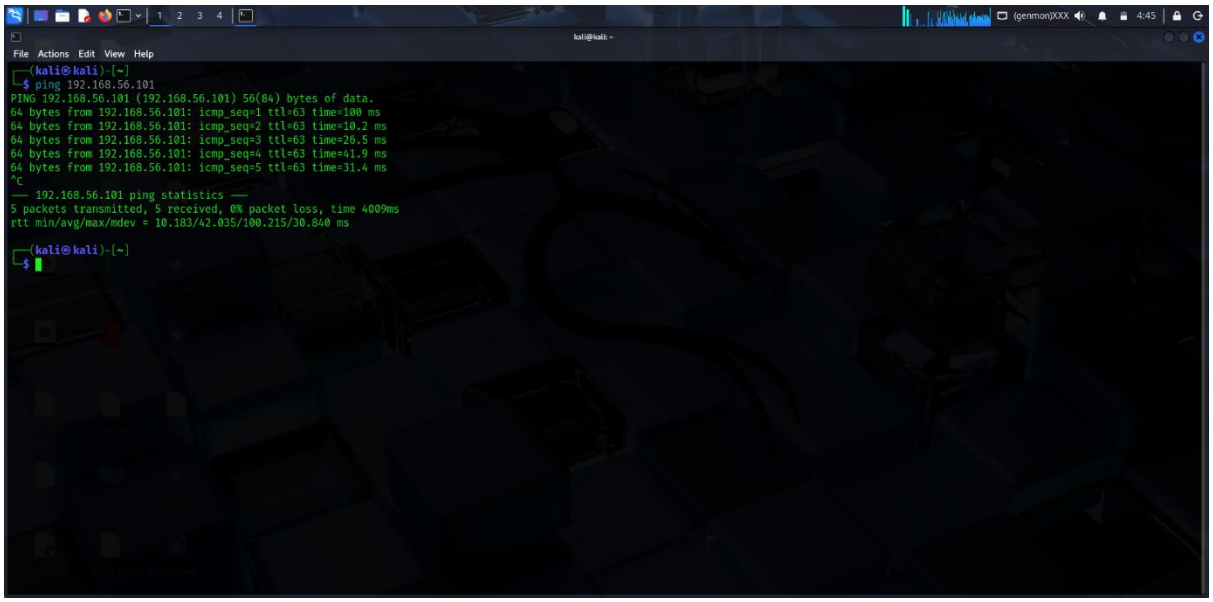
- **Attacker Machine**
  - Operating System: Kali Linux
  - Tool Used: Metasploit Framework
- **Target Machine**
  - Operating System: **Metasploitable2**
  - Target IP Address: **192.168.56.101**

Network Setup
Both machines were connected within a virtual lab environment allowing communication between the attacker and target system.

- **Verify Target Connectivity:** Before launching any exploitation attempt, the connectivity between the attacker machine and the target machine must be verified.

Command Used

- **ping 192.168.56.101:** The ping command confirmed that the Metasploitable2 machine was reachable from the Kali Linux system

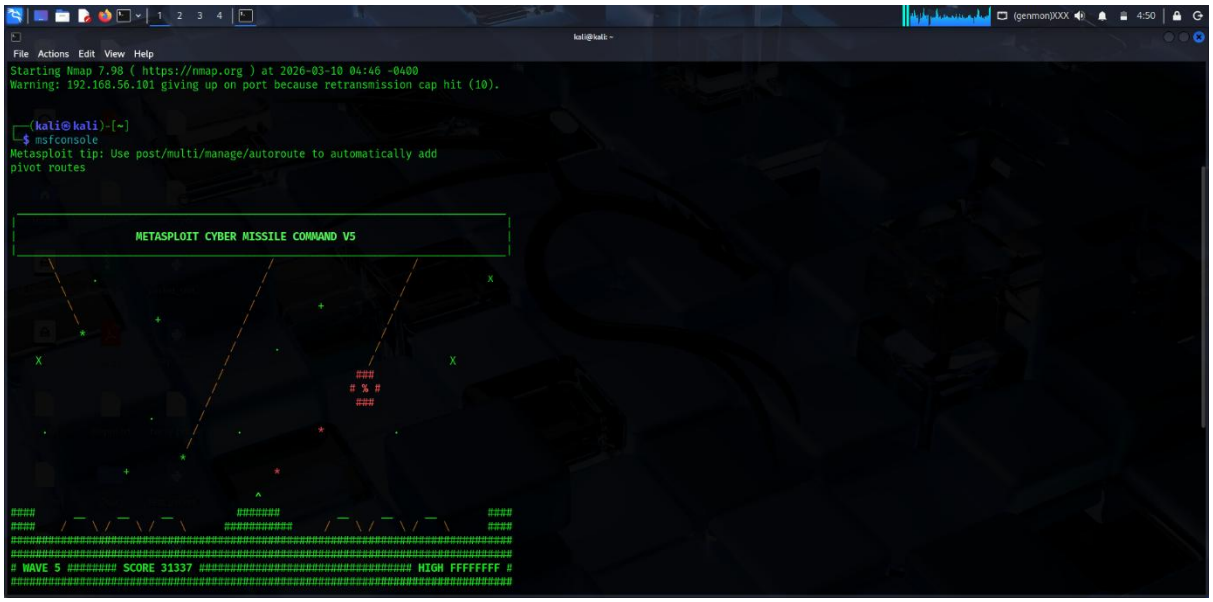


Connectivity verification between Kali Linux and Metasploitable2 using ping command.

- **Launch Metasploit Framework:** Metasploit Framework was started to perform the exploitation activities.

Command Used

- **msfconsole:** Metasploit provides a large database of exploits, payloads, and auxiliary modules used for penetration testing.



Metasploit Framework console interface showing available modules and environment initialization.

➢ Search for Vulnerable Exploit

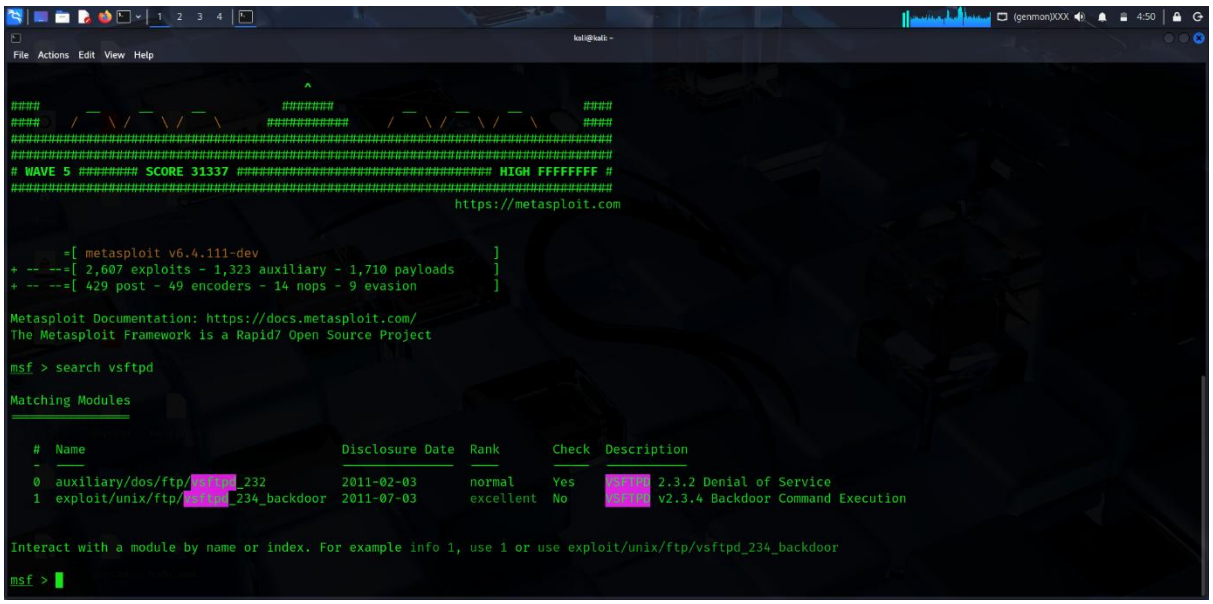
The next step involved searching for known exploits related to the **VSFTPD service** running on the target system.

Command Used

- **search vsftpd:** Metasploit returned multiple modules including

Module	Description
auxiliary/dos/ftp/vsftpd_232	Denial of Service vulnerability
exploit/unix/ftp/vsftpd_234_backdoor	Remote command execution via backdoor

The **vsftpd_234_backdoor exploit** was selected because Metasploitable2 contains this known vulnerable service.



Metasploit search results showing VSFTPD exploit modules.

- **Select the Exploit Module**

The exploit module for the vulnerable FTP service was loaded into the Metasploit environment.

Command Used

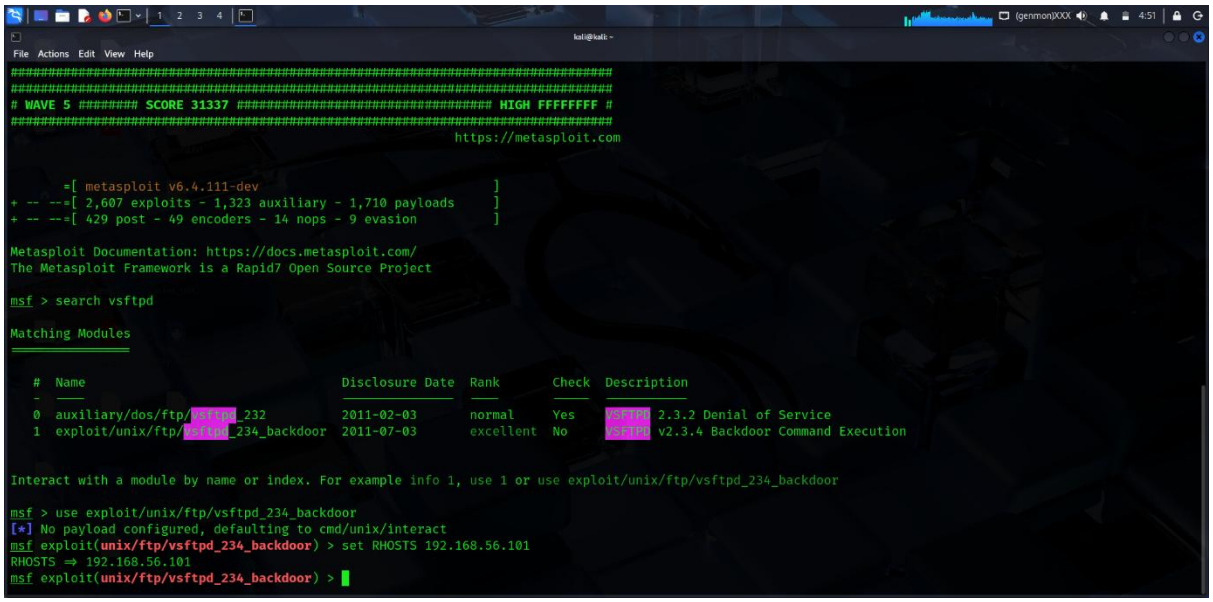
- **use exploit/unix/ftp/vsftpd_234_backdoor:** This module exploits a malicious backdoor inserted in the VSFTPD 2.3.4 service that allows attackers to gain shell access.

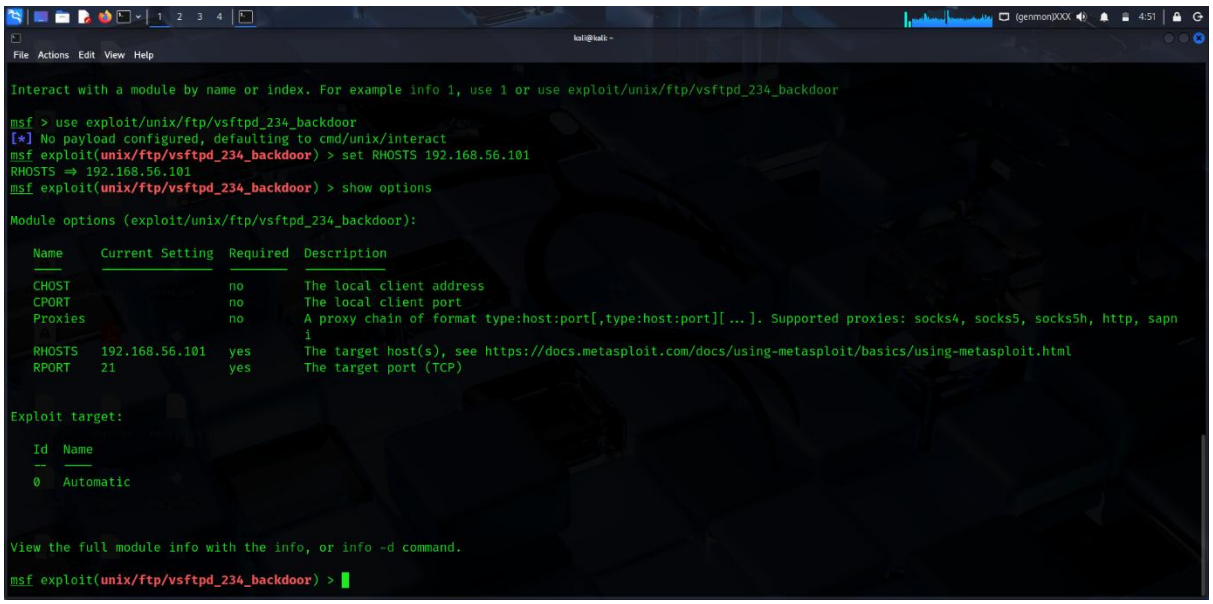
➢ Configure Target IP Address

The target system IP address was configured in the exploit module.

Command Used

- **set RHOSTS 192.168.56.101:** The target system IP address was configured in the exploit module.





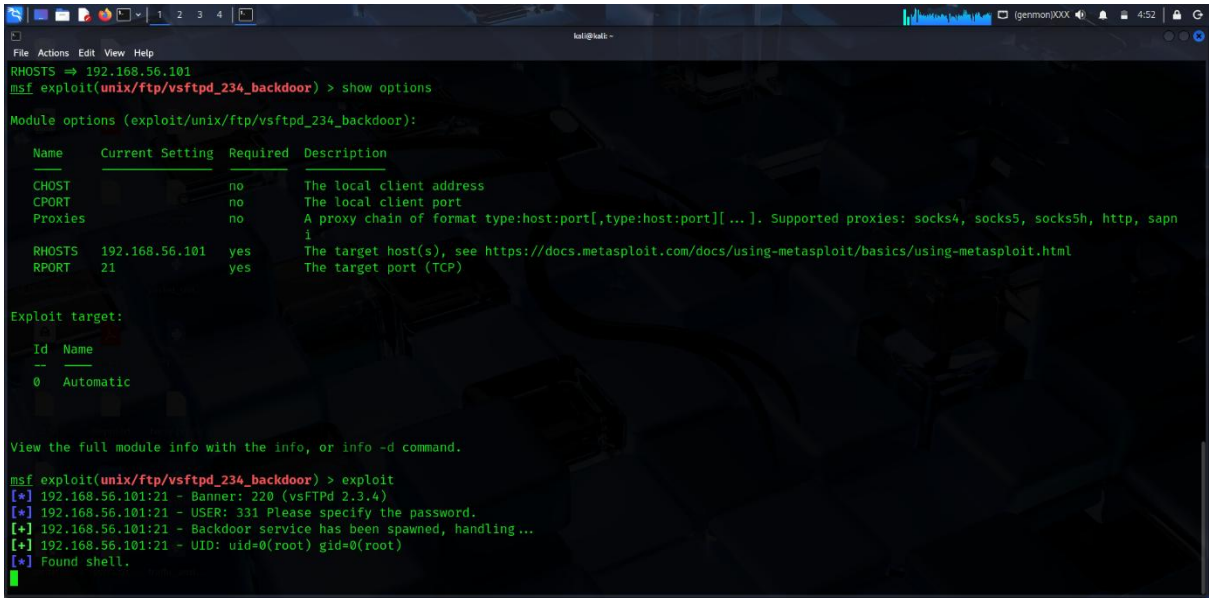
Metasploit exploit configuration showing the target IP address.

- **Execute the Exploit**

The exploit was executed to attempt remote command execution on the vulnerable FTP service.

Command Used

- **Exploit:** The exploit successfully triggered the backdoor in the VSFTPD service and spawned a shell session.



Successful exploitation of VSFTPD backdoor vulnerability and shell access obtained.

Output highlights included

- VSFTPD 2.3.4 service detected
- Backdoor service spawned
- Root privileges obtained
- Command shell session opened

Successful exploitation of VSFTPD backdoor vulnerability and shell access obtained.

Exploit Log Table

Exploit ID	Description	Target IP	Status	Payload
003	VSFTPD 2.3.4 Backdoor Exploit	192.168.56.101	Success	Command Shell

- **Vulnerability Explanation:** The **VSFTPD 2.3.4 backdoor vulnerability** allows attackers to execute arbitrary commands on a target system by exploiting a maliciously modified FTP server version. When triggered, the backdoor opens a command shell that allows unauthorized remote access to the system.

- **Validation Using Exploit-DB:** The vulnerability used in this exploitation phase was validated using the Exploit-DB database. Exploit-DB contains publicly available proof-of-concept exploit codes and vulnerability references used by security researchers and penetration testers.

Exploit-DB confirms that VSFTPD version 2.3.4 contains a malicious backdoor vulnerability that allows remote command execution. Attackers can exploit this flaw to gain shell access to the target system. This vulnerability demonstrates the importance of maintaining updated software and removing outdated services from production environments.

✓ **Security Recommendations**

To prevent exploitation of similar vulnerabilities, the following security practices should be implemented:

1. Always update software to the latest secure versions.
2. Remove or disable unused network services.
3. Monitor FTP services for unusual activity.
4. Implement network intrusion detection systems.
5. Restrict access to critical services using firewall rules.

Conclusion

The exploitation lab successfully demonstrated how attackers can exploit known vulnerabilities in outdated software services. Using the Metasploit Framework, the VSFTPD backdoor vulnerability was exploited to gain command shell access to the target system. This exercise highlights the importance of vulnerability management and regular patching to secure systems against exploitation.